

AUV Acoustic Communication

Project report — Chirp-FSK acoustic link

Overview

A robust acoustic communication concept for an autonomous underwater vehicle operating where GPS is unavailable.

Technical Approach

- Chirp-based signalling for time-frequency robustness
- FSK modulation and demodulation
- DSP filtering and detection
- Embedded C implementation considerations

Outcome

A practical communication architecture balancing range, bandwidth and detection reliability.